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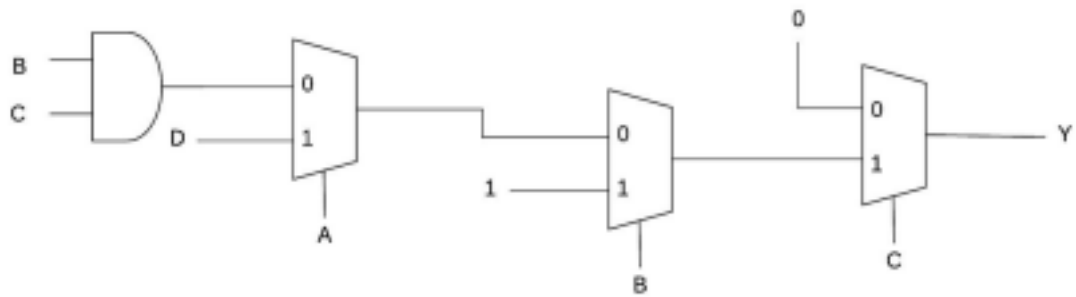
Section	
Seat No.	

CSE 140 Midterm 3
Prof. Tajana Simunic Rosing
Winter 2017

- Do not start the exam until you are told.
 - Write your name and PID at the top of every page. Do not separate the pages. •
- Turn off and put away all your electronics.
- This is a closed-book, closed-notes, no calculator exam.
 - You may only refer to one 8 ½ x 11" page of your handwritten notes.
 - Do not look at anyone else's exam. Do not talk to anyone but an exam proctor during the exam.
 - If you have a question, raise your hand and an exam proctor will come to you.
 - You have 80 minutes to finish the exam. When the time is finished, you must stop. •
- Write your answers in the space provided.
- To get the most partial credit, clearly and neatly show all steps of your work.

Problem 1	16 points	
Problem 2	12 points	
Problem 3	20 points	
Problem 4	15 points	
Problem 5	17 points	
Problem 6	20 points	
Total points	100 points	

1. (16 points) Consider the following circuit using 2:1 multiplexers and an AND gate.

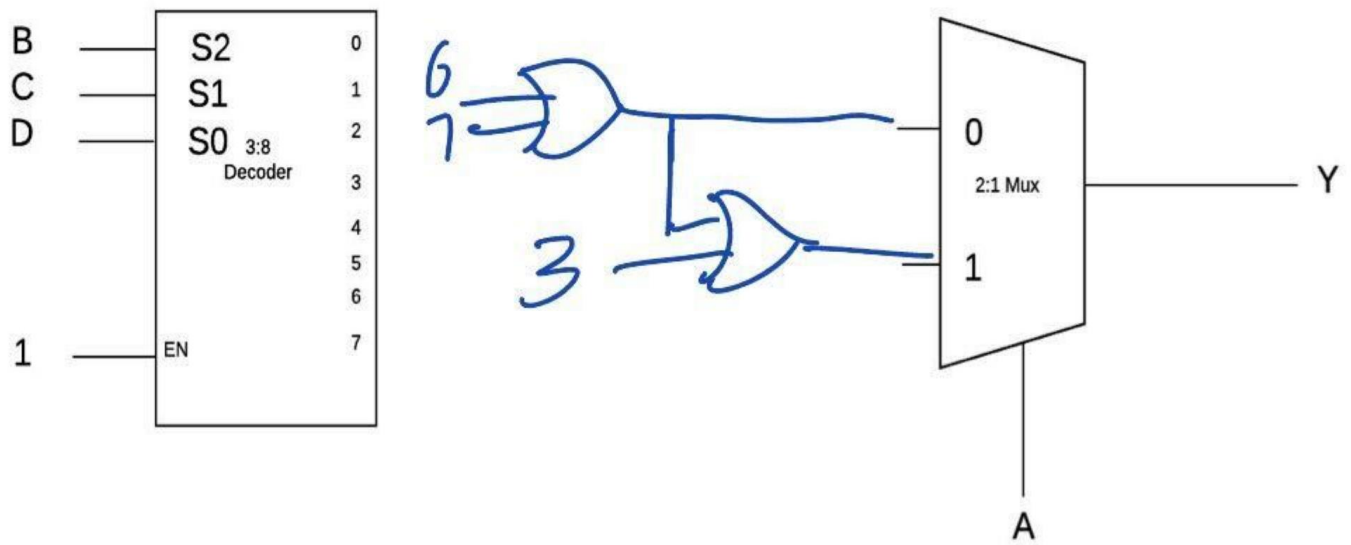


a. Fill out the truth table for Y based on the above circuit.

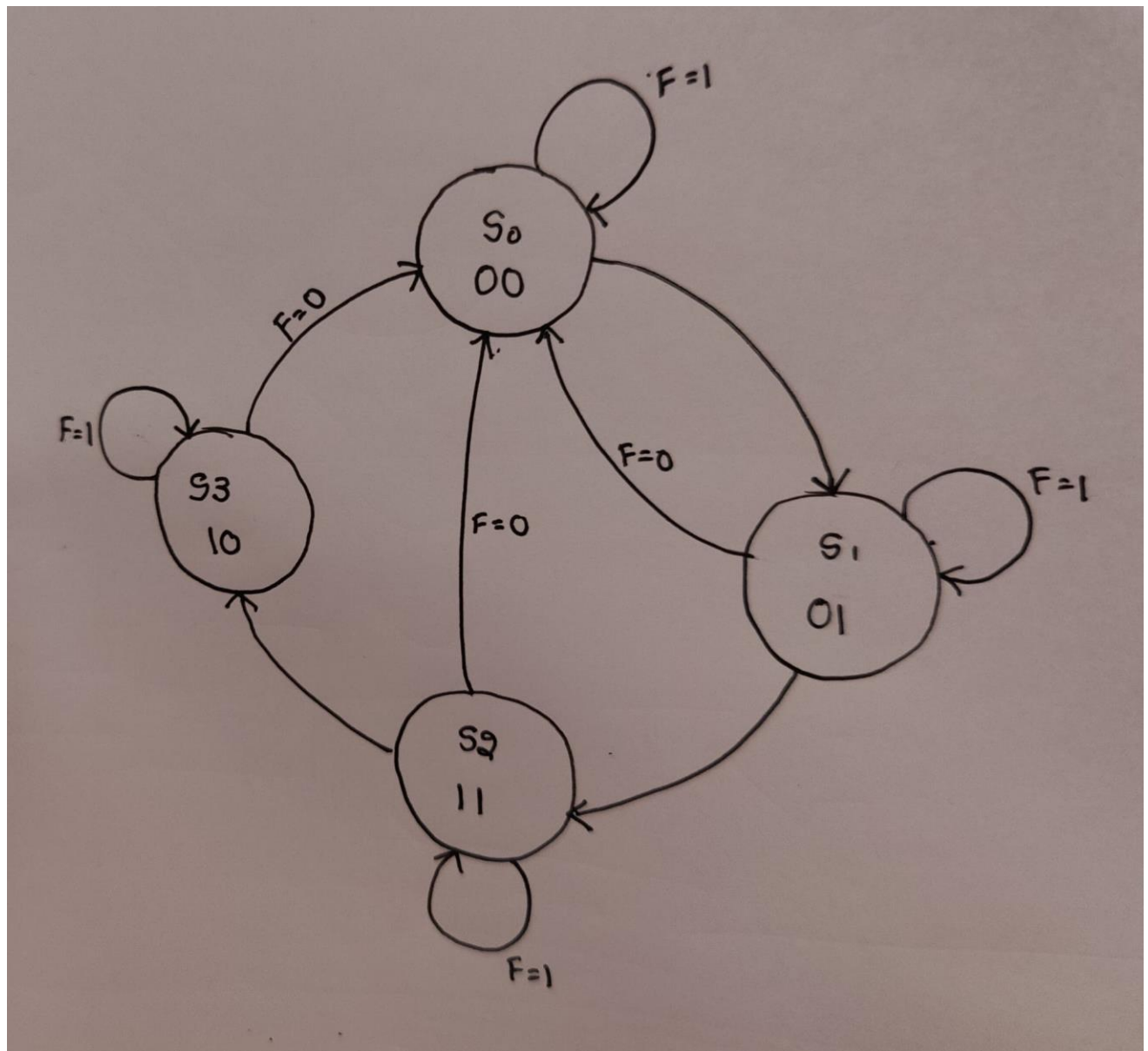
A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0

1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

b. Implement the function represented by the truth table in part a using the incomplete diagram given below with a **minimum** number of multiple-input gates.



2. (12 pts) Consider a Gray code counter which outputs a 2-bit number Y . Y goes from $00 \rightarrow 01 \rightarrow 11 \rightarrow 10 \rightarrow 00$ & repeats. The counter also has an input F which freezes the sequence at its current output when $F = 1$. When $F = 0$, the counter starts again from 00 . Draw an FSM of the counter with minimum number of states.



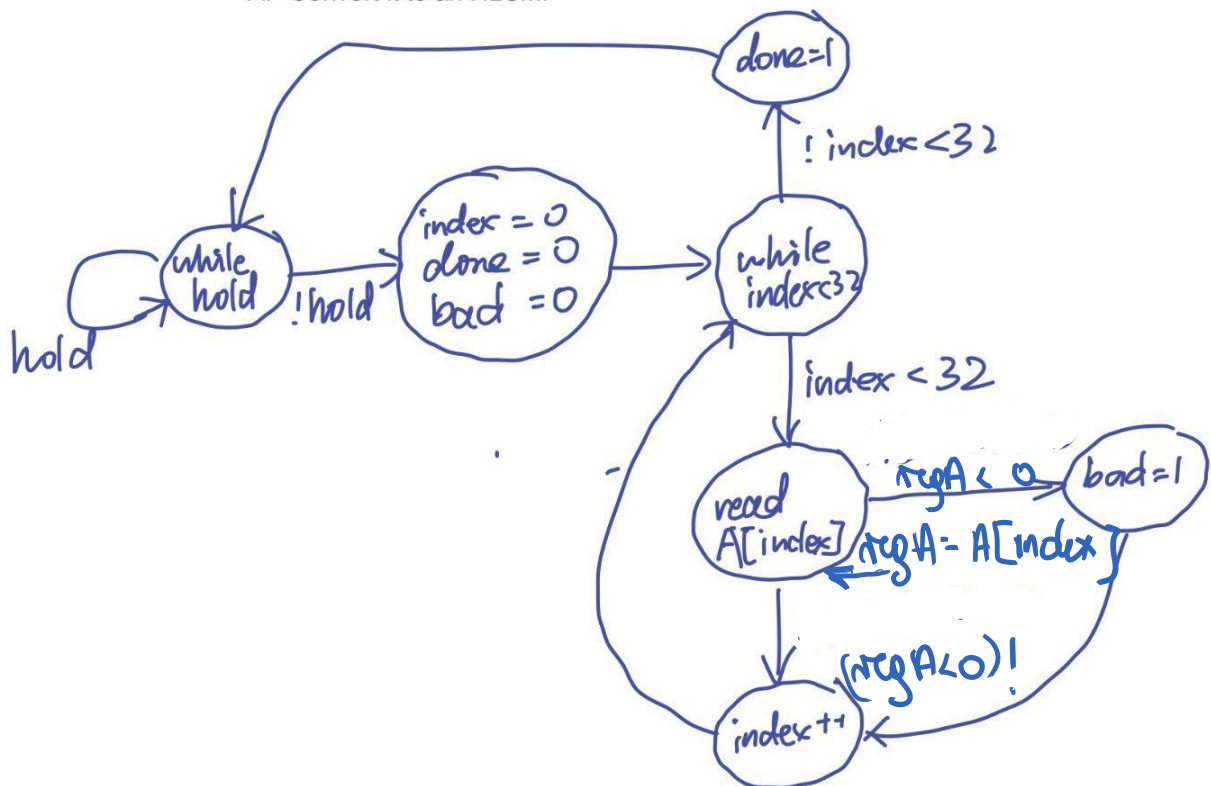
3. (20 pts) Consider the following code.

```

Inputs: byte A[32], bit hold
Outputs: bit done, bit bad
Storage: byte index
main ()
while(1){
    While (hold);
    index = 0;
    done = 0;
    bad = 0;
    while(index<32){
        if (A[index] < 0) {
            bad = 1;
        }
        index = index + 1;
    }
    done = 1;
}

```

A. Convert it to an HLSM.

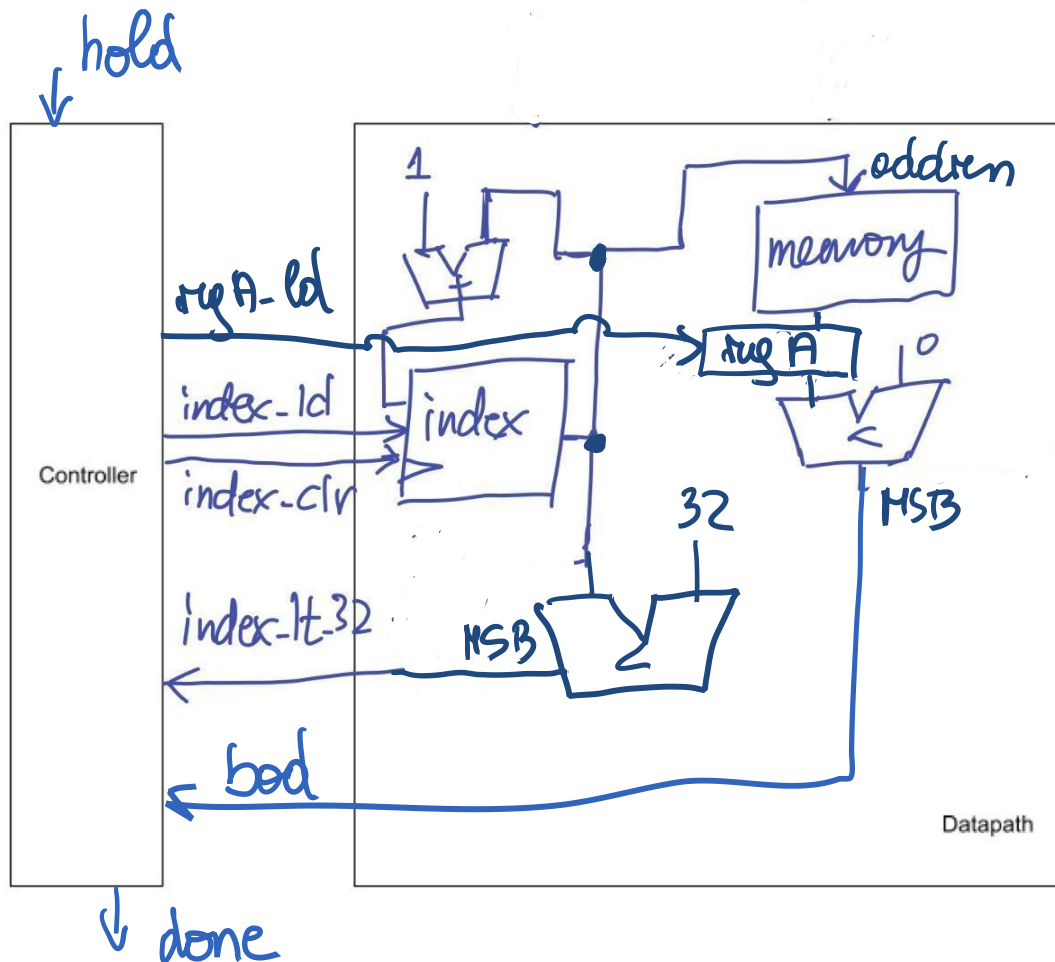


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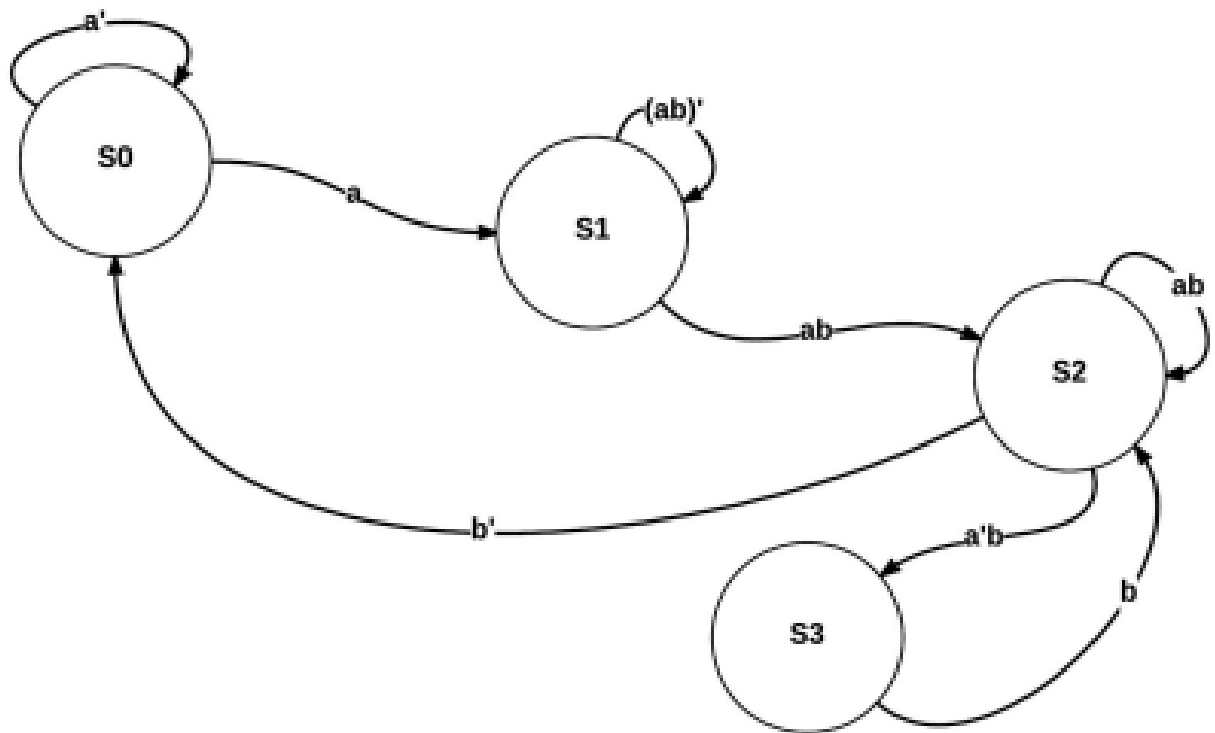
B. Draw the datapath in the diagram below.



C. Show connections between datapath and controller in the diagram above. Label all the signals.

See above

4. (15 Points) Consider the following Finite State Machine.



- a. An engineer has come up with the bit vectors shown in Table 1 for the state assignment. Specify in Table 2 what is the total number of bit changes corresponding to this state assignment.

Table 1	
State	Code
S0	00
S1	11
S2	10
S3	01

Table 2	
Transition	# Bit changes
S0 → S1	2
S1 → S2	1
S2 → S0	1
S2 → S3	2
S3 → S2	2
SUM =	8

b.

State	Q1	Q0	a	b	N1	N0
S0	0	0	0	0	0	0
	0	0	0	1	0	0
	0	0	1	0	1	1
	0	0	1	1	1	1
S3	0	1	0	0	X	X
	0	1	0	1	1	0
	0	1	1	0	X	X
	0	1	1	1	1	0
S2	1	0	0	0	0	0
	1	0	0	1	0	1
	1	0	1	0	0	0
	1	0	1	1	1	0
S1	1	1	0	0	1	1
	1	1	0	1	1	1
	1	1	1	0	1	1
	1	1	1	1	1	0

c. Specify the minimal Boolean expression for MSB of the next state.

ab \ Q1Q0 →	00	01	11	10
00	0	X	1	0
01	0	1	1	0
11	1	1	1	1
10	1	X	1	0

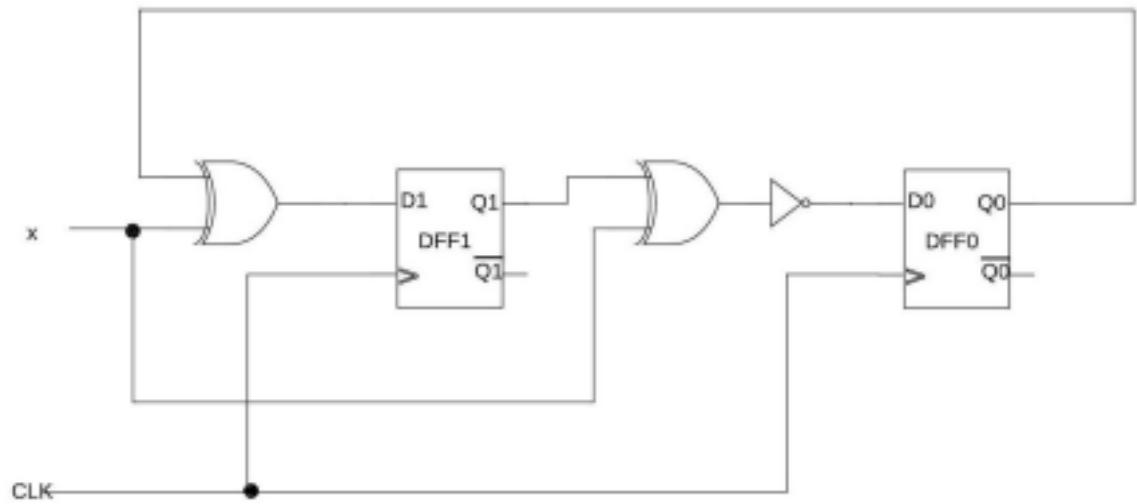
Answer: $Q0 + Q1'Q0'a + ab$

d. Reassign the state assignment so that the number of bit changes is minimized during the state transitions. Complete Tables 3 & 4.

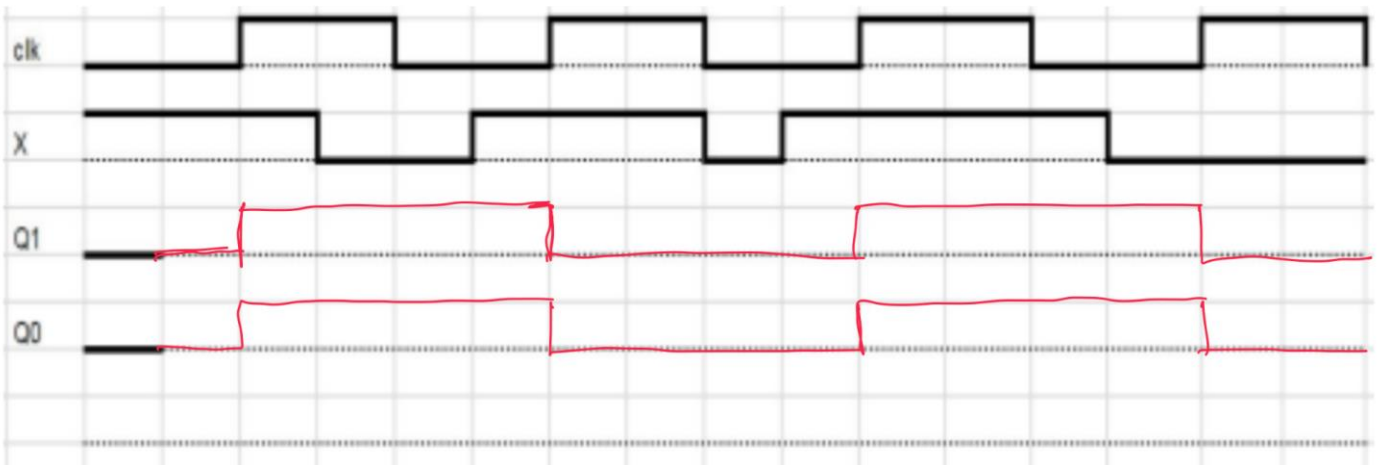
Table 3		
	0	1
0	S0	S3
1	S1	S2

Table 4	
Transition	# Bit changes
S0 → S1	1
S1 → S2	1
S2 → S0	2
S2 → S3	1
S3 → S2	1
SUM =	6

5. (17 points) Consider the circuit shown below.



- a. Complete the waveforms assuming near zero delays and setup and hold constraints equal to 0.



b. Use the following data to answer the questions below

Logic gate	t_{pd}	t_{cd}
NOT	10ns	10ns
XOR	30ns	30ns

t_{setup}	35ns
t_{hold}	25ns
$t_{ccq} = t_{pcq}$	20ns
t_{skew}	5ns

i. Find the maximum clock frequency using the table above.

$$T_C \geq T_{pcq} + T_{setup} + T_{pd} + T_{skew}$$

$$20 + 35 + (30 + 10) + 5$$

$$T_C \geq 100$$

$$f = \frac{1}{100} = 0.01 \text{ Hz}$$

ii. Is there a hold violation? Show your work.

$$T_{ccq} + T_{cd} \geq T_{hold}$$

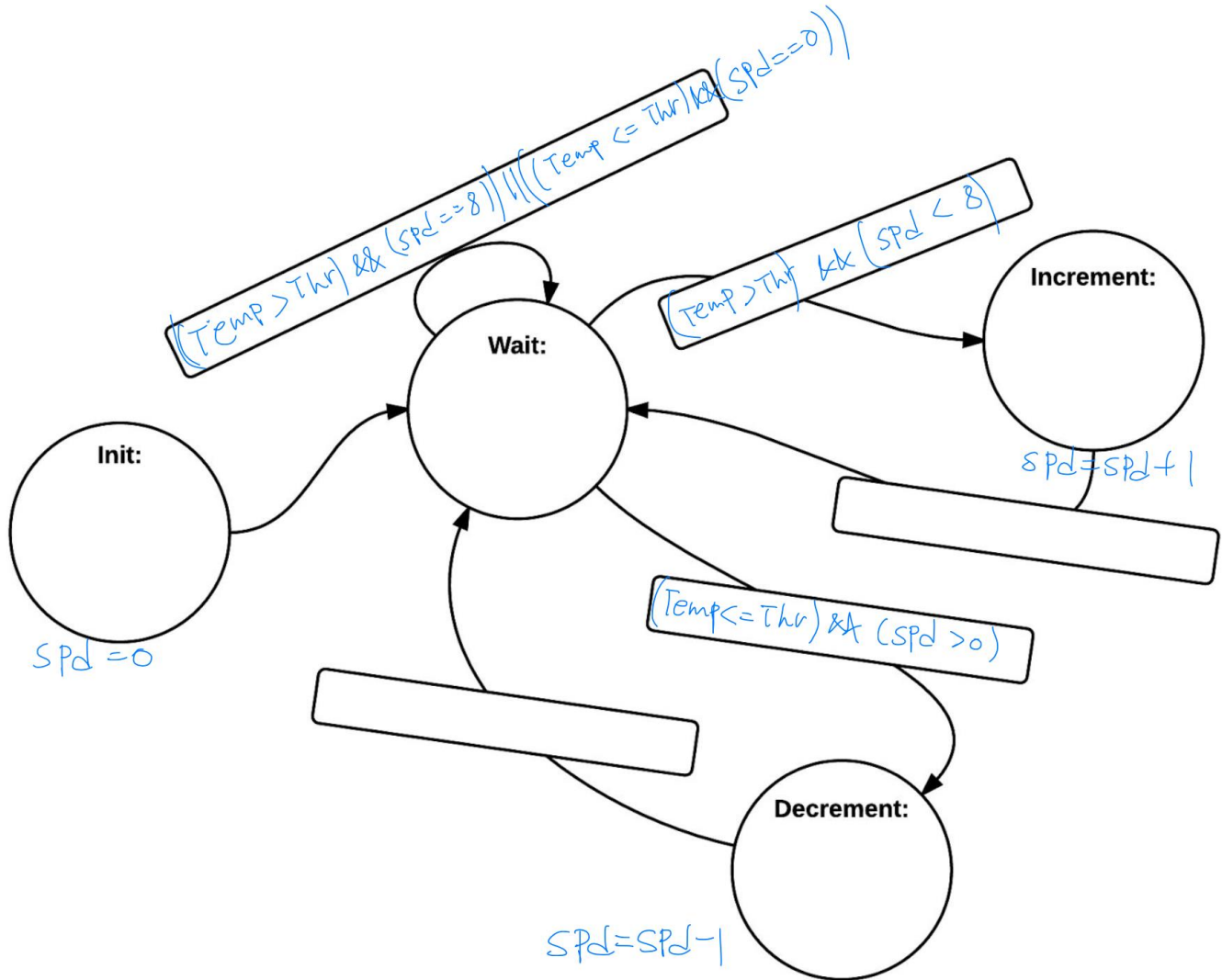
$$20 + 30 \geq 25 \checkmark$$

$$20 + 40 \geq 25 \checkmark$$

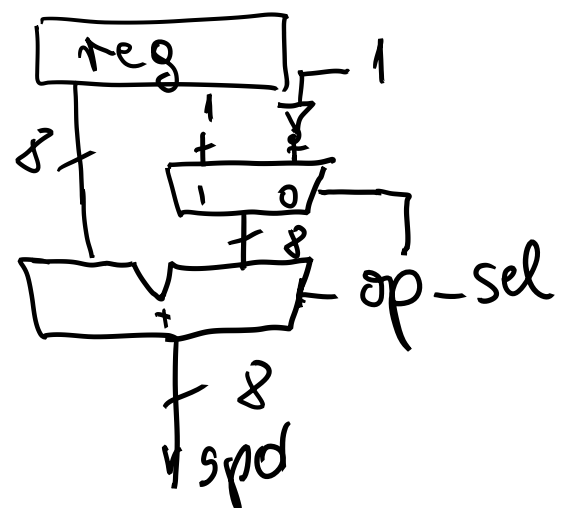
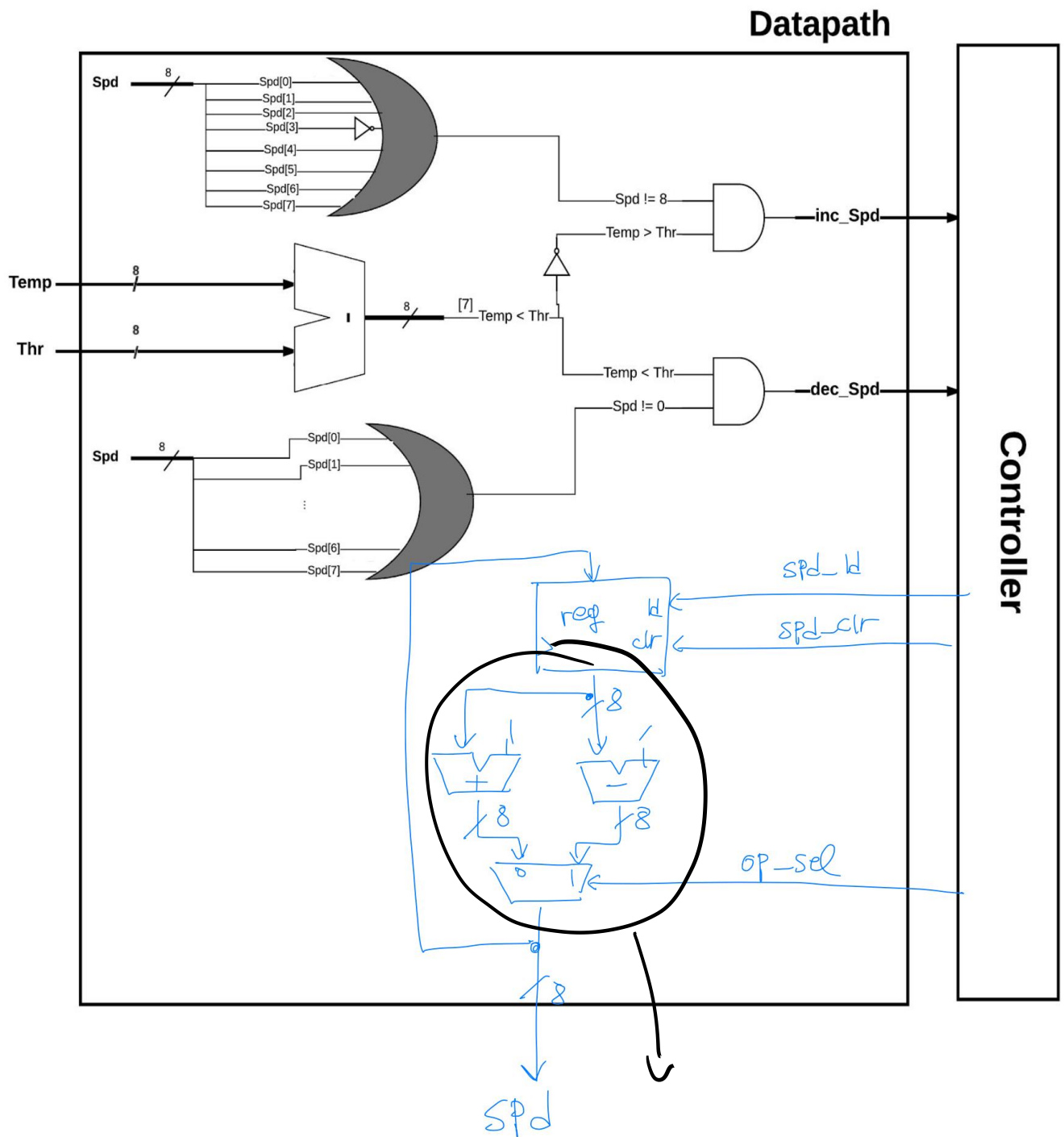
There are no hold violations.

6. (20 Points) Design a cooling system for a furnace. The system input is an 8-bit temperature threshold (Thr) and 8-bit temperature sensor reading (Temp). The system outputs an 8-bit number (Spd) which is speed of a digitally controlled fan. Spd can only take values from 0 to 8. If Temp is below or equal to Thr, fan speed (Spd) is decremented. If Temp is above Thr, fan speed is incremented.

a. Fill in the missing information (transition conditions and state arithmetic operations) on the HLSM of this system.



b. Complete the datapath for this system and connect the controller to datapath



c. Draw the controller FSM.

